A COHERENT MODULE ON A RINGED SPACE IS OF FINITE PRESENTATION AND HENCE QUASI-COHERENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The two clauses of the definition of a coherent module: local generation by finitely many sections and local generation of the kernel of the resulting surjection. The author's section on coherent modules, in which the counted statement occurs, specifically the definition immediately preceding it. Supplies both inputs of the local presentation constructed in the proof.
- The notion of a module of finite presentation and its consequence of being quasi-coherent. The author's section on modules of finite presentation, invoked in the statement itself. Identifies the local presentations produced by the proof with the conclusion of the statement.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Any coherent $\mathcal{O}_X$-module is of finite presentation and hence quasi-coherent.*

Proof. Let $\mathcal{F}$ be a coherent sheaf on X. Pick a point $x \in X$. By (1) of the definition of coherent, we may find an open neighbourhood U and sections s_i, $i = 1, \ldots, n$ of $\mathcal{F}$ over U such that $\Psi : \bigoplus_{i=1,\ldots,n} \mathcal{O}_U \to \mathcal{F}$ is surjective. By (2) of the definition of coherent, we may find an open neighbourhood V, $x \in V \subset U$ and sections $t_1, \ldots, t_m$ of $\bigoplus_{i=1,\ldots,n} \mathcal{O}_V$ which generate the kernel of $\Psi|_V$. Then over V we get the presentation

$$\bigoplus\nolimits_{j=1,\ldots,m} \mathcal{O}_V \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} \mathcal{O}_V \to \mathcal{F}|_V \to 0$$

as desired. □

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